

Tide retrospective: the normalized singular identifiability theorem

germbij Question 7.7 answered, by integration by parts

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Abstract

Two nonnegative analytic losses whose Laplace families agree *projectively* beyond all orders, $\int \phi e^{-tL_2} - C(t) \int \phi e^{-tL_1} = o(t^{-\infty})$ for every $\phi \in C_c^\infty$ and some scalar $C(t)$, coincide on a neighbourhood of every common zero. This is the normalized form of the singular identifiability problem of the germbij note (“What expectation values know about the loss landscape”): with $C = Z_2/Z_1$ the hypothesis is exactly equality of the normalized expansions $\Phi_{L_1} = \Phi_{L_2}$, so the note’s last open question closes with no anchoring point required. The mechanism is a three-line integration by parts that a co-author added to the note on 2026-08-26 but that had no theorem environment and no Lean: $\int \partial_v \phi e^{-tL} = t \int \phi \partial_v L e^{-tL}$, so the observables $\partial_v \phi$ and $\phi \partial_v L_1$ cancel the unknown scalar exactly and leave a plain family beyond all orders, which the existing sector lower bound turns into vanishing of the germ of $\partial_v(L_2 - L_1)$. Five hundred lines of new Lean in one file, no `sorry`, no axiom, four LSP rounds, one strategic consult; the scalar C may be an arbitrary function. The corollary with the $1/Z$ incorporated is stated separately.

1 Setting

The germbij note asks when the family of posterior expectation values $\langle \phi \rangle_t = \int \phi e^{-tL}/Z(t)$, read as asymptotic expansions in t , determines the germ of the loss L at its minimum. The laplace seabed carries the note’s earlier answers end to end: the nondegenerate theorem (jet and germ recovery modulo a constant, with the location), the flat-perturbation counterexamples, and the singular theorem `pencil_families_force_eq_near_smooth`, which is the *unnormalized* statement, agreement on the nose of $\int \phi e^{-tL_1}$ and $\int \phi e^{-tL_2}$. The note’s Question 7.7 was whether the normalized data, which only fix the unnormalized families up to a common scalar series $C(t)$, still force equality; the seabed had the one-point anchoring wrapper `one_point_anchoring_contradiction`, which closes the gap wherever the zero locus has a point of constructive recovery, and nothing in general.

Surveying the paper for this tide showed that the question is no longer open on paper: the Overleaf commit of 2026-08-26 inserted a proof by integration by parts directly under the Question, without a theorem environment, without a Lean counterpart, and with the surrounding prose still describing the pre-proof state. The user’s direction was to continue the germbij formalisation and to consider what the newly formalised `hironaka` and `greybook` seabeds unlock. The deliberation

(tide log tide-log/2026-09-19-06-05-tide-germbij-normalized.md, consult in gpt-germbij-normalized.v1.md) weighed three candidates: the normalized theorem; a cross-seabed forward direction (leading Θ -asymptotics of $\int e^{-tK}$ from hironaka’s sublevel-volume statement plus an Abelian transfer); and the projective form of “equal zero loci are forced”. Both parties voted for the normalized theorem: one coherent statement, every ingredient present in the seabed, and it is the paper’s headline change since August. The cross-seabed bridge was recorded as the next arc (it needs a toolchain bump of laplace to v4.33.1, two new dependencies, and a Θ -version of the greybook transfer that does not yet exist).

2 The thing formalised

Let $L_1, L_2 : \mathbb{R}^d \rightarrow \mathbb{R}$ be smooth and nonnegative, analytic at a common zero p . Suppose there is a function $C : \mathbb{R} \rightarrow \mathbb{R}$ such that for every smooth compactly supported ϕ ,

$$\int \phi e^{-tL_2} dw - C(t) \int \phi e^{-tL_1} dw = o(t^{-N}) \quad \text{for every } N.$$

Then $L_1 = L_2$ on a neighbourhood of p . If this holds at every point of a set W_0 of common zeros, $L_1 = L_2$ on an open neighbourhood of W_0 . In particular, if the expectation values normalized by a common nonnegative window χ agree beyond all orders for every ϕ , the same conclusion holds.

The Lean signature of the point form:

```
theorem normalized_families_force_germ_eq_at
  {L1 L2 : (iota -> R) -> R} {p : iota -> R}
  (h1 : ContDiff R infty L1) (h2 : ContDiff R infty L2)
  (hL1 : forall w, 0 <= L1 w) (hL2 : forall w, 0 <= L2 w)
  (hA1 : AnalyticAt R L1 p) (hA2 : AnalyticAt R L2 p)
  (hp1 : L1 p = 0) (hp2 : L2 p = 0) {C : R -> R}
  (hfam : forall phi : (iota -> R) -> R,
    ContDiff R infty phi -> HasCompactSupport phi ->
    SuperPoly fun t => (integral w, phi w * Real.exp (-(t * L2 w)))
      - C t * integral w, phi w * Real.exp (-(t * L1 w))) :
  forallf w in nhds p, L1 w = L2 w
```

with the seabed’s “beyond all orders” predicate

```
SuperPoly f := forall N : Nat, f =o[atTop] fun t => t ^ (-(N : R))
```

The locus form `normalized_families_force_eq_near` quantifies the hypotheses over $p \in W_0$; the form with the normalization, `normalized_expectations_force_eq_near`, takes a continuous, compactly supported, nonnegative window χ with $Z_i(t) = \int \chi e^{-tL_i} > 0$ and the hypothesis $\int \phi e^{-tL_2}/Z_2 - \int \phi e^{-tL_1}/Z_1$ beyond all orders.

The hypotheses are what the argument uses and no more. Smoothness of L_1 everywhere is what makes $\phi \partial_v L_1$ an admissible observable (analyticity near W_0 would do, with observables

localized to that neighbourhood; the global form was kept because it removes support book-keeping). Analyticity enters only through finite vanishing order of the amplitude $\partial_v(L_2 - L_1)$, exactly as in the unnormalized theorem. Nothing at all is assumed about C : not positivity, not a power-log expansion, not measurability. Compactness of W_0 is not used; the neighbourhood is the union of the per-point ones.

3 Proof strategy

Write $I_i(\phi, t) = \int \phi e^{-tL_i}$ and $R_\phi = I_2(\phi) - C I_1(\phi)$; the hypothesis is that every admissible R_ϕ is beyond all orders. For a direction v , integrating by parts against the Boltzmann weight gives $I_i(\partial_v \phi) = t I_i(\phi \partial_v L_i)$. Hence, for $t > 0$,

$$\frac{R_{\partial_v \phi}(t)}{t} - R_{\phi \partial_v L_1}(t) = I_2(\phi \partial_v(L_2 - L_1), t),$$

an exact identity in which C has cancelled. Both observables on the left are smooth with compact support, so the right side is beyond all orders for every ϕ ; in the file this is `superPoly_integral_fderiv_sub_mul_exp`. This is the whole idea; the rest is the seabed.

Set $g = L_2 - L_1$ and $a = \partial_v g$. Taking $\phi = \psi a$ with ψ a smooth bump at p turns the family into $\int \psi a^2 e^{-tL_2}$. If the germ of a at p were nonzero, the sector lower bound would give $\int \psi a^2 e^{-tL_2} \geq \kappa t^{-m-d/2}$ for large t , m the vanishing order of a ; this is the content of the new packaged lemma `analytic_square_weight_eq_zero_near`, which translates p to the origin, extracts the least nonzero diagonal of the power series of a (`exists_least_nonzero_diagonal`), the Taylor remainder (`analytic_remainder_bound`) and the scaled set (`leading_part_scaled_set`), the quadratic upper bound on the weight $K = L_2$ (`quadratic_upper_bound_of_nonneg`, which needs only C^2 , nonnegativity and $K(p) = 0$), a `ContDiffBump` equal to 1 on the ball where that bound holds, and then `sector_lower_bound_multi` on the window $(\sqrt{t})^{-1}S$ where the bump is 1, compared with the full integral by `setIntegral_le_integral`; the contradiction is `lower_bound_not_superpolynomial`. The one hypothesis of the diagonal lemma that is not free is $a(p) = 0$, and it holds because both losses have a local minimum at p , so $dL_1(p) = dL_2(p) = 0$ (`IsLocalMin.fderiv_eq_zero`).

So $\partial_v g = 0$ near p for every coordinate direction v ; over the finitely many directions this gives $dg = 0$ on a ball around p , g is constant on the ball by convexity (`Convex.is_const_of_fderivWithin_eq_zero`, only C^1 used), and $g(p) = 0$. The locus form is the union of the balls, copied from the pencil theorem. The corollary with the normalization takes $C = Z_2/Z_1$ and uses $0 \leq Z_2(t) \leq \int \chi$ for $t \geq 0$ to pass from the normalized difference to the projective one (`SuperPoly.bounded_mul`).

4 Roadblocks and resolutions

There were none of substance; this was a four-round file. The rounds were mechanical and are worth recording because they will recur.

Grades. On this pin `ContDiff.differentiable` wants $\infty \neq 0$ and `ContDiff.fderiv_right` wants $\infty + 1 \leq \infty$, neither of which `le_top` proves (the top of `WithTop Nat-infty` is ω , not

∞); by `simp` and `contDiff_infty_iff_fderiv` discharge them. For C^2 at an analytic point, `AnalyticAt.contDiffAt` gives any grade for free, where `ContDiffAt.of_le` against ∞ needs a coercion lemma one has to hunt for.

Derivative of a difference. `fderiv_sub` is stated for `Pi` subtraction and does not rewrite `fun w => L2 w - L1 w`; `(hasFDerivAt.sub hasFDerivAt).fderiv` does.

The clash the per-file check cannot see. The file compiled clean under the LSP with its own `SuperPoly.sub` and `SuperPoly.congr`, and the library root then failed: both already exist, in `CutoffRemoval` and `LocatedCutoff`, files the new one did not import. The fix was to import them and delete the duplicates; the seabed's `congr` takes `f =f[atTop] g` and hands the pointwise goal as an applied lambda, so a `beta_reduce` was needed before `congr 1`. Lesson: grep the library for a lemma name before declaring it, even in a leaf file.

One consult, before writing. The deliberation consult vetted the inserted proof and shaped the file. Its load-bearing paragraph:

GPT-6 Astra: Consequently, for $t > 0$, $R_{D\phi}(t)/t - R_{\phi DL_1}(t) = I_2(\phi D(L_2 - L_1), t)$. This is the clean cancellation identity to formalise. Its right-hand side is superpolynomially small because that class is closed under subtraction and division by t . There is no multiplication of an error by C .

It also flagged that the amplitude vanishes at p automatically, that local constancy must be stated on a ball rather than globally, and that C needs no property at all, all of which the file follows. The only divergence from the deliberation plan is that the square-weight rigidity was written as a standalone lemma with the family hypothesis quantified over all smooth bumps, rather than inlined; this is what let the main theorem stay at forty lines.

5 What was Mathlib, what was new

Mathlib provided the integration by parts,

$$\text{integral_mul_fderiv_eq_neg_fderiv_mul_of_integrable}$$

with $f = \phi$ and $g = e^{-tL}$, whose three integrability side conditions come from compact support via the `integrable_of_hasCompactSupport` lemma for continuous functions; the derivative of the exponential (`HasFDerivAt.exp`), the compact-support book-keeping (`HasCompactSupport.fderiv`, `.comp_left`, `.mul_right`), the bump functions (`ContDiffBump`), the convexity argument for local constancy, and the asymptotic algebra (`IsLittle0.mul_isBig0`, `IsBig0.mul_isLittle0`, `isBig0.const_of_tendsto`).

The seabed provided everything analytic: the least nonzero diagonal, the remainder bound, the scaled set, the quadratic upper bound, the sector lower bound and the decay contradiction, plus the `SuperPoly` predicate and its closure under subtraction and congruence.

New in this tide: the directional integration-by-parts identity for Boltzmann weights (`integral_fderiv_mul_exp`), the cancellation lemma, the square-weight rigidity lemma (the first reusable form of the sector bound for an amplitude that is not the pencil difference), the finite-direction local-constancy lemma, two closure lemmas for `SuperPoly` (division by

t , multiplication by a bounded function), and the three theorems. About 520 lines, of which the rigidity lemma is roughly a third.

6 Lessons

- For `CLAUDE.md`: the three grade and derivative gotchas above (`infty != 0` via `by simp`; `fderiv_sub` is Π -shaped; `AnalyticAt.contDiffAt` for any finite grade), and “grep for the name before declaring a `SuperPoly.*` helper”.
- For the tide skill: the survey step paid for itself. The candidate came from reading the Overleaf history, not the Lean; a 33-line insertion without a theorem environment is invisible to anything that greps for `\begin{theorem}`. When a paper is the seabed’s source, diff the paper against the last pinned commit before drafting candidates.
- For the formalisation skill: a per-file LSP check is not a library build. The clash surfaced only when `Laplace.lean` imported both files; `lake build` of the root before the first commit is the gate, as the seabed’s `CLAUDE.md` already says.

7 Follow-ups

- *Scalar rigidity*. After $L_1 = L_2$ near p , a bump ψ_0 there has $I_1(\psi_0) = I_2(\psi_0)$ exactly, so $(1 - C(t))I_1(\psi_0)$ is beyond all orders and $I_1(\psi_0) \gtrsim t^{-d/2}$ gives `SuperPoly (fun t => C t - 1)`. Needs the Gaussian-type lower bound $\int \psi_0 e^{-tL_1} \gtrsim t^{-d/2}$ from the quadratic upper bound, which the seabed states as a hypothesis (`hanchor_low`) but does not yet prove.
- *Equal zero loci, projective form*. A bump at $p \in \{L_1 = 0\} \setminus \{L_2 = 0\}$ forces $C = o(t^{-\infty})$ and then every $I_2(\phi)$ beyond all orders, impossible if L_2 has a zero; the reverse inclusion needs $|C(t)| = O(t^{d/2})$ from a bump at a zero of L_1 . Elementary, no analyticity; would let the theorem’s “common zeros” read “zeros of L_1 ”.
- *Support-local hypotheses*. Replace global smoothness by analyticity on a neighbourhood of W_0 with observables supported there, using `contDiff_mul_of_tsupport_subset`.
- *Cross-seabed arc*. A standalone Θ -Abelian transfer ($\mu\{K \leq \varepsilon\} = \Theta(\varepsilon^\lambda(\log 1/\varepsilon)^m)$ implies $\int e^{-tK} d\mu = \Theta(t^{-\lambda}(\log t)^m)$), which needs no hironaka dependency; then, after bumping laplace to v4.33.1, compose with hironaka’s sublevel-volume theorem `exists_hasLLCExponentsOn_of_analyticOnNhd_nonneg` for the leading term of the singular expansion of an analytic loss.
- *Paper*. The note should promote the inserted proof to a theorem (statement drafted in the Overleaf working file `germbij_slop.tex`, section S1), delete the “we expect the answer is no” paragraph, demote Proposition 7.8 to a constructive remark, and carry `\leanref` markers for the three theorems once the merge SHA is pinned. Paper edits await the author’s direction.

Acknowledgements

The mechanism is the co-author's integration-by-parts argument in the note. GPT-6 Astra vetted it and fixed the shape of the cancellation identity and the rigidity lemma. The seabed is the germbij singular arc of 2026-08-09 and 2026-08-10, in particular [the singular-smooth tide](#) whose bump construction this file copies.